Seungju Han

seungjoohan1@gmail.com | 609-558-8851 | New York, NY | LinkedIn | Github | Portfolio

Aspiring Machine Learning Engineer with over 3 years of experience in real-time pose estimation, on-device model optimization, and full-stack ML pipeline development.

PROFESSIONAL EXPERIENCE

ViFiVE, New York, NY — Aug 2022 – May 2025
Data Scientist
- Deployed and optimized machine learning models for on-device production in iOS and web environments, ensuring high performance and reliability.
- Boosted **3D pose estimation accuracy by 17%** with a custom self-attention module, significantly improving clinical assessment performance.
- Developed a comprehensive ML pipeline using MySQL, Firestore, and Python, **boosting clinical assessment efficiency by 30%** across multiple tests.
- Optimized data augmentation pipeline and training workflows, **reducing computation overhead by 10%.**
- **Enhanced a Random Forest pose classification model accuracy from 89% to 96%** by applying advanced feature engineering techniques, boosting reliability of clinical assessment predictions.
- Led efforts to validate the proficiency of AI-based models in high-stakes clinical assessments with Stanford and Seoul National University Hospital, **impacting over 300 patients and clinicians** across different regions.

New York Mets, Remote — Oct 2021 – Jul 2022
Data Scientist
- Generated expected distribution model for infield and outfield batted balls with 89% accuracy by leveraging pitcher and batter data, including pitch types, pitcher/batter handedness, and batted ball statistics.
- Applied clustering (K-Means, KNN) to group similar pitchers and simulated realistic pitcher-batter matchups, **enhancing player performance forecast by 25%.**
- Informed defensive alignment strategies using model insights, contributing to the Mets **improving their Defensive Runs Saved (DRS) by 130%**, from +20 DRS in 2021 to +46 DRS in 2022.

Fastcampus Language, Seoul, Korea — Jun 2020 – Aug 2020
Business Analyst
- Automated KPI dashboard for leadership team, **reducing operating time by 75%** through integrating Google Suite tools (BigQuery, Looker, and Google Sheets).
- Leveraged SQL to analyze data for multiple teams, uncovering key trends and providing actionable insights that improved decision-making and contributed to a **13% decrease in operating cost on average**.

PROJECTS

Pose Human Pose Estimation with Vision Transformers
Github: github.com/seungjoohan/dino_pose
- Built 3D human pose estimator using Vision Transformer models, fine-tuned with Low-Rank Adaptation (LoRA).
- Developed a data processing pipeline for handling pose datasets and integrated augmentation strategies.

Scalable Movie Rating Prediction
- Developed a scalable end-to-end data pipeline to preprocess over 28 million movie and user records using Spark and stored engineered features in MongoDB for efficient retrieval.
- Trained an Alternative Least Square (ALS) collaborative filtering model, achieving r2 score of 0.83 within 2 hours using distributed machine learning, leveraging the scalability of Spark to process the large dataset efficiently.

EDUCATION

University of San Francisco, San Francisco, CA — Jul 2022
MS in Data Science
Relevant Coursework: Machine Learning, Deep Learning, Linear Regression, Time Series Analysis, A/B Testing, Relational Databases, NoSQL, Distributed Computing (Spark), Data Structures & Algorithms

University of California Berkeley, Berkeley, CA — Dec 2019
BA in Statistics

TECHNICAL SKILLS

Python (numpy, pandas), Machine Learning (pytorch, tensorflow, scikit-learn), Deep Learning, SQL (BigQuery, Postgres, MySQL), NoSQL (MongoDB, Firestore), Spark, Data Visualization (matplotlib, Seaborn), AWS (S3, EC2, EMR), Databricks, GCP, Git